

Let G be a finite group. ~~k a field~~

①.

Let R be a commutative unital ring (In particular, field)

RG : a free R -module with basis G

$$x = \sum_{s \in G} x_s s \quad \text{and} \quad y = \sum_{t \in G} y_t t \quad \text{in } RG$$

$$xy = \sum_{s, t \in G} x_s y_t st = \sum_{g \in G} \left(\sum_{s \in G} x_s y_{s^{-1}g} \right) g \quad \text{take } R := k, \text{ a field.}$$

Let $\tau: kG \rightarrow k$ be the linear form on kG defined by

$$\tau(g) = 0 \text{ if } g \neq 1 \text{ and } \tau(1) = 1$$

Lemma: (i) For all $x, y \in kG$, $\tau(xy) = \tau(yx)$ (class function)

(ii) The map $\hat{\tau}: kG \rightarrow (kG)^* = \text{Hom}_k(kG, k)$

$$x \mapsto (y \mapsto \tau(xy))$$

is a k -linear isomorphism.

A class function: $f: G \rightarrow k$ s.t. $f(g^{-1}xg) = f(x)$

$\forall g, x \in G$, which implies $f(xg) = f(gx)$

extend linearly to $kG \rightarrow k$

It shows kG is a symmetric algebra.

$$\text{Identify } \left\{ \begin{array}{c} \text{functions } G \rightarrow k \\ \parallel \\ F(G, k) \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{linear form of } kG \\ \parallel \\ \text{Hom}_k(kG, k) = (kG)^* \end{array} \right\}$$

For $f \in F(G, k)$, the inverse image of f under

$$\hat{\tau} \text{ is } f^\circ = \sum_{g \in G} f(g^{-1})g$$

here $CF(G, k)$ denote the space of class function on G

Lemma: Let $a \in kG$, $f \in F(G, k)$. Denote af the element of $F(G, k)$ defined by $(af)(x) = f(xa)$

$\forall a \in kG$. Then $(af)^{\circ} = af^{\circ}$

Let $\mathcal{C}(G)$ be the set of conjugacy classes of G

For $C \in \mathcal{C}(G)$ set

- γ_C the characteristic function on C , i.e

$$\gamma_C(g) = \begin{cases} 0 & \text{if } g \notin C \\ 1 & \text{if } g \in C \end{cases}$$

- $\hat{C} = \sum_{g \in C} g \in kG$

For a class function f we have

$$f = \sum_{\substack{g \in C \\ C \in \mathcal{C}(G)}} f(g_C) \gamma_C, \quad f^{\circ} = \sum_{C \in \mathcal{C}(G)} f(g_C^{-1}) \hat{C}, \quad g_C \in C$$

It implies γ_C is a basis of class functions.

Lemma: Let $Z(kG)$ denote the center of the group algebra kG , $Z(kG) = \{x \in kG \mid xy = yx, \forall y \in kG\}$

(1) The family $\{\gamma_C\}_{C \in \mathcal{C}(G)}$ is a basis of $CF(G, k)$

and the family $\{\hat{C}\}_{C \in \mathcal{C}(G)}$ is a basis of $Z(kG)$

Hence $\dim_k CF(G, k) = |\mathcal{C}(G)|$

(2). The isomorphism $\hat{\cdot}$ resorives ~~to~~ to a k -linear

isomorphism $CF(G, k) \rightarrow Z(kG)$. Hence $\dim_k Z(kG) = |\mathcal{C}(G)|$

(M, ρ) a k -representation of G . The morphism $\rho: G \rightarrow GL(M)$ extends by linearity to an morphism

$$kG \rightarrow \text{End}_k(M)$$

• Give an algebra morphism $\rho: kG \rightarrow \text{End}_k(M)$ sending 1 to 1

it restricts to a group morphism $G \rightarrow GL(M)$

• Given an algebra morphism $\rho: kG \rightarrow \text{End}_k(M)$

define $kG \times M \rightarrow M$

$$(x, m) \mapsto \rho(x)m$$

M ~~becomes~~ becomes a kG -module.

Hence $\text{Rep}(G, \text{vect}_k) \hookrightarrow kG\text{-mod}$.

Lemma: Let $\rho: kG \rightarrow \text{End}_k(kG)$ be algebra morphism associated with regular representation.

then ρ is injective. i.e. regular representations are faithful, $kG(kG)$ is faithful.

Lemma. (Schur) Let (S, ρ) and (S', ρ') be two irreducible representations of G .

1°. If (S, ρ) and (S', ρ') are not isomorphic, then

$$\text{Hom}_{kG}(S, S') = 0$$

2°. $\text{End}_{kG}(S) = \text{Hom}_{kG}(S, S)$ is a division ~~ring~~ k -algebra finite dimensional over k

Q2 If $k = \bar{k}$, the $\text{End}_{kG}(S) = (k \cdot \text{Id}_S)$

Ex. $\rho: G \rightarrow GL_n(k)$ be an irreducible representation of G ,
If $M \in GL_n(k)$ s.t. $M^{-1} \rho(g) M = \rho(g) \quad \forall g \in G$ then $k = \bar{k}$
 M is scalar.

Theorem: If $\text{char } k \mid |G|$, then kG is not semisimple.

Theorem: If $\text{char } k \nmid |G|$, then kG is semisimple.

proof: Let V be a kG -module W be any kG -submod of V

Aim: show W has complement

Let $\iota: W \rightarrow V$ be the inclusion, ι is a kG -morphism

As k -spaces. $V = W \oplus W'$ for some k -space of W'

Let $p: V \rightarrow W$ be the projection. $\leadsto p \circ \iota = \text{id}_W$

consider $\pi: V \rightarrow W$

$$v \mapsto \frac{1}{|G|} \sum_{g \in G} (\rho(g)^{-1} \circ p \circ \rho(g)) v$$

$$\forall h \in G, \rho(h)^{-1} \pi \rho(h) = \frac{1}{|G|} \sum_{g \in G} \rho(gh)^{-1} p \rho(gh) = \pi$$

$\Rightarrow \pi$ is a kG -morphism

$$\bullet \pi \circ \iota = \frac{1}{|G|} \sum_{g \in G} \rho(g)^{-1} p \rho(g) = \text{id}_W.$$

Let $U = \ker \pi \Rightarrow U$ is a kG -module and $V = W \oplus U$ as kG -module.

Def. If a field k is a splitting field for kG , then we say k is a splitting field for G .

Theorem (Branes) If k contains a $|G|$ -root of unity
Then k is a splitting field for G .

prop: $\text{char } k = p$ G : a p -group. Then the only simple kG -mod is the trivial module (kG is projective over of simple module).

pf: let S be a simple module.

let $0 \neq x \in S$, let $X_x = \{gx \mid g \in G\} \subseteq S$.

S is abelian group.

let S_0 be the subgroup of S generated by X
 G acts on S , and S_0 is invariant under the action of G

S_0 is abelian p -group. $\rightarrow G$ acts on S_0

a p -grp acts on a p -grp.

S_0 except 0 , G has another fixed points in S_0

$\exists y \in S_0$ s.t $y \neq 0$, $gy = y, \forall g \in G$.

So y generates a kG -submod of S

G acts trivially on y

By simplicity, $S \cong \langle y \rangle \sim$ trivial module.

Theorem: $\text{char } k \nmid |G|$ ($|G|$ is invertible in k)

(1). As a ring, kG is a direct sum of matrix alg. over division rings

(2). Assume further k is a splitting field of G (In particular, $k = \bar{k}$)

let $S_1, \dots, S_r$ be ^{all} pairwise non-isomorphic simple kG -modules.

let $d_i = \dim_k S_i$. Then $kG \cong S_1^{d_1} \oplus \dots \oplus S_r^{d_r}$

and $|G| = d_1^2 + \dots + d_r^2$

$$kG \cong M_{d_1}(k) \oplus \dots \oplus M_{d_r}(k)$$

(M, ρ) k -representation. $\rho: G \rightarrow GL(M)$

Take a basis of $M \leadsto G \rightarrow GL_d(k)$, $d = \dim_k M$

Let $k = \mathbb{C}$

Def: Given a $\mathbb{C}$ -representation of G , $\rho: G \rightarrow GL(\mathbb{C}^{\check{M}})$

define a function $\chi_V: G \rightarrow \mathbb{C}$, $g \mapsto \text{tr } \rho(g)$

χ_V is called the character of (V, ρ)

prop: (1) χ_V is a class function

$$(2). \chi_V(g^{-1}) = \overline{\chi_V(g)} \quad \forall g \in G.$$

(3) If $V \cong W$ as $\mathbb{C}G$ -modules, then $\chi_V = \chi_W$

$$(4). \chi_{V \oplus W} = \chi_V + \chi_W, \quad \chi_{V \otimes W} = \chi_V \chi_W$$

$$\overline{\chi_{V^*}(g)} = \chi_V(g) = \chi_V(g^{-1}) = \overline{\chi_V(g)} \quad \text{i.e. } \chi_{V^*} = \overline{\chi_V}$$

$$\chi_{\text{Hom}_{\mathbb{C}}(V, W)}(g) = \chi_V(g^{-1}) \chi_W(g) \quad \text{i.e. } \chi_{\text{Hom}_{\mathbb{C}}(V, W)} = \overline{\chi_V} \chi_W.$$

$$(5). \chi_V(1) = \dim V$$

$|\chi_V(g)| \leq \dim V$ with equality iff the action of g on

V is a scalar, $|\chi_V(g)| = \dim V$ iff g acts trivially on V .

$CF(G) := CF(G, \mathbb{C})$ for short

For $f_1, f_2 \in CF(G)$, define $\langle f_1, f_2 \rangle = \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)}$

$$\langle f_1, f_2 \rangle = \frac{1}{|G|} \sum_{c \in \text{cl}(G)} \frac{|G|}{|G(c)|} f_1(g_c) \overline{f_2(g_c)} = \sum_{c \in \text{cl}(G)} \frac{1}{|G(c)|} f_1(g_c) \overline{f_2(g_c)}$$

prop: $\langle, \rangle$ is a Hermitian form of the $\mathbb{C}$ -space $CF(G)$

If f_1 and f_2 are characters of $\mathbb{C}$ -representation of G .

$$\begin{aligned} \text{then } \langle f_1, f_2 \rangle &= \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g^{-1})} = \frac{1}{|G|} \sum_{g \in G} f_1(g^{-1}) \overline{f_2(g)} \\ &= \frac{1}{|G|} \sum_{g \in G} f_2(g) \overline{f_1(g)} = \langle f_2, f_1 \rangle \end{aligned}$$

$$\Rightarrow \langle f_1, f_2 \rangle \in \mathbb{R}$$

lemma: Let V be a $\mathbb{C}G$ -module. $V^G = \{v \in V \mid gv = v, \forall g \in G\}$

Then V^G is a $\mathbb{C}G$ -submodule of V and.

$$\dim_{\mathbb{C}} V^G = \langle \chi_V, 1_G \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_V(g)$$

proof: Let $\pi: V \rightarrow V$ $v \mapsto \left(\frac{1}{|G|} \sum_{g \in G} g \right) v$

Then $\pi^2 = \pi$, $\text{Im } \pi = V^G$

$$\bullet \quad \pi(hv) = \frac{1}{|G|} \sum_{g \in G} (gh)v = \pi(v) = \frac{1}{|G|} \sum_{g \in G} hg v = h \pi(v), \quad \forall h \in G, v \in V$$

$\Rightarrow \pi$ is a $\mathbb{C}G$ -morphism.

$$\pi^2 = \left(\frac{1}{|G|} \sum_{g \in G} g \right) \left(\frac{1}{|G|} \sum_{h \in G} h \right) = \frac{1}{|G|^2} \sum_{g, h \in G} gh = \pi.$$

$$\text{If } v \in V^G \text{ then } \pi(v) = \frac{1}{|G|} \sum_{g \in G} gv = v \Rightarrow v \in \text{Im } \pi$$

$$\forall h \in G, h \left(\frac{1}{|G|} \sum_{g \in G} gv \right) = \frac{1}{|G|} \sum_{g \in G} hgv \Rightarrow \text{Im } \pi \subseteq V^G.$$

$$\Rightarrow V = \ker \pi \oplus \text{Im } \pi, \quad \pi|_{\text{Im } \pi} = \text{id}_{\text{Im } \pi}$$

$$\text{trace}(\pi) = \dim_{\mathbb{C}} V^G$$

$$\text{trace}\left(\frac{1}{|G|} \sum_{g \in G} g\right) = \frac{1}{|G|} \sum_{g \in G} \text{trace}(p(g)) = \frac{1}{|G|} \sum_{g \in G} \chi_V(g)$$

Lemma: k any field, $V, W : kG$ -mod. $\langle \chi_V, 1 \rangle = \langle \chi_V, 1 \rangle$

$$\text{Hom}_k(V, W)^G = \text{Hom}_{kG}(V, W)$$

Theorem: (Row orthogonality relations).

V, W simple GG -modules. Then $\langle \chi_V, \chi_W \rangle = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{otherwise} \end{cases}$

$$\text{P.f. } \dim_{\mathbb{C}} \text{Hom}_{\mathbb{C}G}(V, W) = \dim_{\mathbb{C}} (\text{Hom}_{\mathbb{C}}(V, W)^G) = \langle \chi_{\text{Hom}_{\mathbb{C}}(V, W)}, 1 \rangle$$

$$= \langle \bar{\chi}_V \chi_W, 1 \rangle = \frac{1}{|G|} \sum_{g \in G} \bar{\chi}_V(g) \chi_W(g) = \langle \chi_V, \chi_W \rangle$$

$$\text{Irr}(G) = \{ \text{irreducible characters of } G \}$$

character afforded by simple GG -modules.

Corollary: $\text{Irr}(G)$ is $\mathbb{C}$ -linearly independent.

Corollary: $\text{Irr}(G)$ is a $\mathbb{C}$ -basis of $\mathbb{C}F(G, \mathbb{C})$

$$\text{proof: } |\text{Irr}(G)| = r, \quad \mathbb{C}G = M_{d_1}(\mathbb{C}) \oplus \dots \oplus M_{d_r}(\mathbb{C})$$

$$\mathbb{Z}(\mathbb{C}G) = \mathbb{Z}(M_{d_1}(\mathbb{C}) \oplus \dots \oplus M_{d_r}(\mathbb{C}))$$

$$\dim_{\mathbb{C}} \mathbb{Z}(\mathbb{C}G) = r = |\text{Irr}(G)|$$

$$\dim_{\mathbb{C}} \mathbb{C}F(G)$$

$$\text{Corollary: } |\text{Irr}(G)| = |\mathbb{C}F(G)|$$

$$\text{Lemma: } |G| = \sum_{\chi \in \text{Irr}(G)} \chi(1)^2$$

Corollary: $V: \mathbb{C}G$ -module. $\langle \chi_V, \chi_V \rangle = 1 \Leftrightarrow V$ is simple. (5)

Lemma: $V, W: \mathbb{C}G$ -module

$$V = \sum_1^{n_1} \oplus \dots \oplus \sum_r^{n_r} \quad W = \sum_1^{m_1} \oplus \dots \oplus \sum_r^{m_r} \quad \begin{array}{l} \text{complete.} \\ \text{(rep. set)} \end{array}$$

$\sum_1 \dots \sum_r$ non-isomorphic $\otimes$ simple $\mathbb{C}G$ -modules. $\wedge$

m_i, n_i non-negative integers.

$$\langle \chi_V, \chi_W \rangle = m_1 n_1 + \dots + m_r n_r$$

Cor: $\chi_V = \chi_W$ iff $V \cong W$

Cor: $V: \mathbb{C}G$ -mod. $V \cong \sum_1^{n_1} \oplus \dots \oplus \sum_r^{n_r}$. $\sum_1 \dots \sum_r$ non-iso, sim.

$$n_i \geq 0 \\ \text{then } n_i = \langle \chi_V, \chi_{\sum_i} \rangle$$

Thm (Column, orthogonality relation) Let χ be the character table of G .

$$\chi = \begin{array}{c} \chi_1 \\ \vdots \\ \chi_r \end{array} \quad \begin{array}{c} g_1 \dots g_r \\ \chi_i(g_j) \end{array}$$

$$\text{Ir}(G) = \{\chi_1, \dots, \chi_r\}$$

$$\text{Cl}(G) = \{ \overbrace{g_1 \dots g_r}^{c_1 \dots c_r} \}$$

Proof: row ortho relation $\Rightarrow \chi \underbrace{\left(\frac{1}{|G|} \text{diag}(|G|, \dots, |G|) \right)^{-1}}_{C^{-1}} \bar{\chi}^T = I_r.$

So $\chi C^{-1} \bar{\chi}^T = I \Rightarrow C^{-1} \bar{\chi}^T \chi = I \Rightarrow \bar{\chi}^T \chi = C$

$\Rightarrow \sum_{\chi \in \text{Ir}(G)} \chi(g) \overline{\chi(h)} = \begin{cases} |C_G(g)|, & \text{if } g \text{ and } h \text{ are conjugate in } G. \\ 0 & \text{otherwise} \end{cases}$